OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

The OpenAIRE Graph, explored through Claude

Data model, aggregation workflow, discovery & monitoring

Paolo Manghi

OpenAIRE AMKE, CTO

Institute of Information Science and Technologies, CNR

OpenAIRE

ISTITUTO DI SCIENZA E TECNOLOGIE DELL'INFORMAZIONE "A. FAEDO"

OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

For more: *http://graph.openaire.eu/docs*

THE OPENAIRE GRAPH

OpenAIRE Graph

A Scholarly Knowledge Graph for Open Research Intelligence

Timely, comprehensive, open, community-governed.

Built on open data, open source, and open APIs

Aggregates metadata from 155,000+ data sources

Monthly updates, rigorous deduplication, and AI-powered enrichment

Steered by community requirements, priorities, and decisions

370M+ research publications, data, software

400+ funders, 4mi grants, with links to products

7 billion citation relationships pub-pub, pub-data, pub-sw

Full-text mining of links, AI-based Fields of Science and SDG classification, usage data, APC tracking — the complete scholarly communication record.

Strategic choice stays in Europe’s hands— not with commercial vendors.

KEY PRINCIPLE

Sovereignty over Research Intelligence means controlling how knowledge is represented and interpreted.

THE OPENAIRE GRAPH · DATA MODEL

Six entity types, one interlinked graph

Live counts from the Graph API v3 (July 2026). Every entity carries PIDs and provenance.

Research products

380M+

publications · datasets · software · other

Data sources

155K+

repositories, journals, aggregators

Organizations

497K

institutions & companies (ROR)

Projects

3.9M

grants from 400+ funders

Persons

ORCID

researchers, linked to their outputs

Communities

40+

research infrastructures & initiatives

7 billion citation relationships *(pub–pub, pub–data, pub–software) plus affiliation, funding, versioning and provenance links connect the entities.*

OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

The Aggregation Workflow: from 155K+ sources to one Graph

750Mi+ records

COLLECT

Harvest metadata from 2000+ repositories, journals, Crossref, DataCite, ORCID, ROR, funders

500Mi+ records

TRANSFORM

Map records onto the common data model & vocabularies, resolve DOI clashes

370Mi+ records

DEDUPLICATE

Cluster & merge records of the same product or organization

65Mi+ full-texts

ENRICH

Full-text mining/AI to infer links, FoS & SDG classification, citations, usage data

PUBLISH

Monthly releases — APIs, Zenodo dumps, portals

370Mi+ records
~7Bi citation links

Every month the full pipeline re-runs end-to-end: the Graph is rebuilt, deduplicated, enriched and re-published as a fresh release.

OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

From API calls to conversation

The OpenAIRE MCP server exposes ~28 tools over the Graph API v3 and ScholeXplorer. Claude picks tools, chains calls, and synthesizes — no query syntax needed.

1

You ask

"Top-influence LLM papers since 2020?"

›

2

Claude routes

search_research_products(influence_class=C1, ...)

›

3

Graph responds

structured records + pagination totals

›

4

Claude replies

ranked answer, links, next steps

Same graph, three surfaces: *a chat prompt, a scheduled digest, or a live dashboard artifact.*

DISCOVERY · EXAMPLE 1

“Most influential LLM papers since 2020”

search_research_products(query=“large language models”, influence_class=C1, sort=influence DESC)

Swin Transformer: Hierarchical Vision Transformer ICCV 2021	**19,494** citations
ChatGPT for Good? Opportunities & Challenges for Education Learning & Individual Differences, 2023	**4,526** citations
ChatGPT on USMLE: AI-assisted medical education PLOS Digital Health, 2022	**3,314** citations
Large language models encode clinical knowledge Nature, 2023	**2,794** citations

Influence C1 = top 0.01% of field, time-normalized. *13 matches; top 4 shown. Ranked by the Graph’s bibliometric indicator, not raw counts.*

DISCOVERY · EXAMPLE 2

Follow the citation network of a landmark paper

explore_research_relationships(target_pid=10.1038/s41586-021-03819-2) · AlphaFold, Nature 2021

37,537

incoming citation links
via ScholeXplorer

AlphaFold, Nature 2021 — protein-structure prediction

What Claude concludes from the network

1. **Reach spills far beyond biology**

   Citing work clusters in pharmacology, oncology and drug discovery — a structural-biology result reused as a tool across the life sciences.

2. **It became research infrastructure**

   Citations are dominated by method reuse (e.g. RAF1 kinase, S1P transport in Cell) rather than critique — the hallmark of an enabling technology.

3. **It seeds a new AI-for-science genre**

   Transformer follow-ons like DeepB3P show the paper spawning derivative ML models, not just applications.

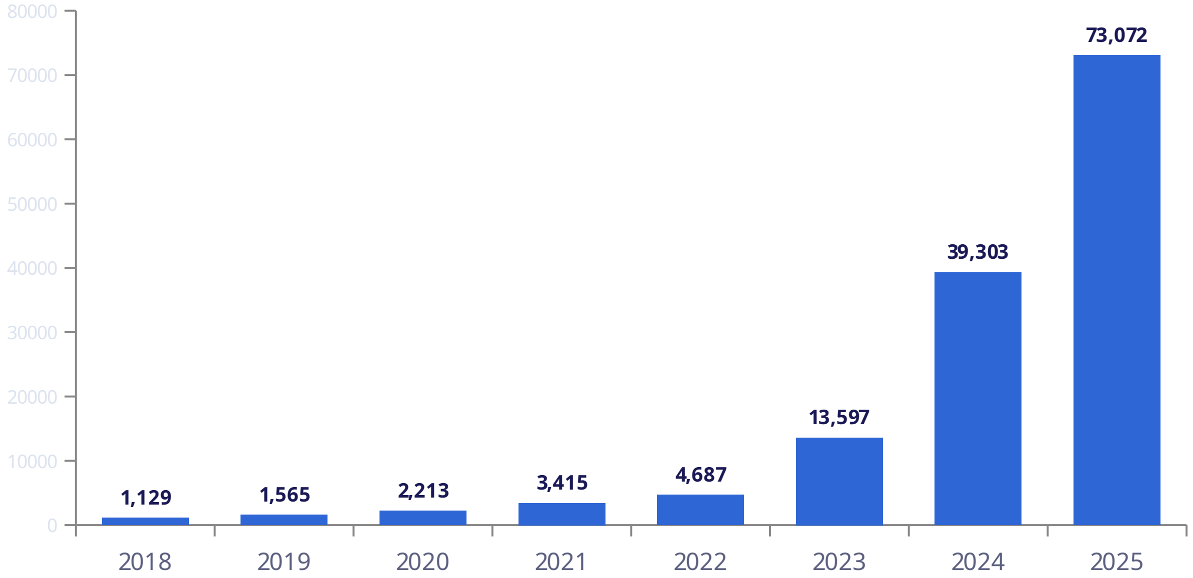
One hop, three findings: *the Graph turns a citation count into an interpretable map of how a landmark result actually gets used.*

OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

MONITORING · EXAMPLE 1

Track a topic's growth over time

analyze_research_trends(search="large language models", 2018–2025)



What Claude flags

65×

growth 2018→2025

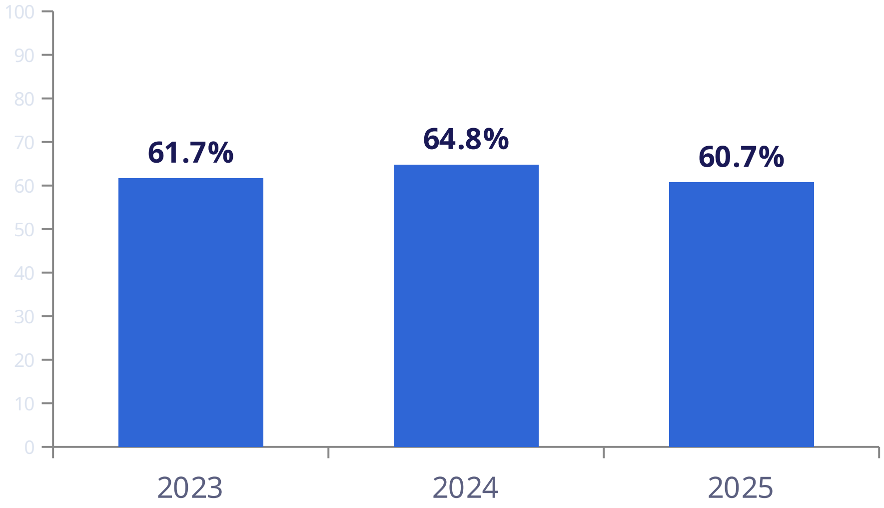
Inflection at 2023 (ChatGPT era): +190% vs 2022, then +170% in 2024.

~139K total products

MONITORING · EXAMPLE 2

Open-science trend for one institution

search_research_products(rel_organization_id=University of Padua) + open-access & type filters, per year



Open-access share of annual output

What Claude flags for Padua

1. **~62% open access, and holding**

   Of ~17K products/yr, well over half are open — 61.7% → 64.8% → 60.7%. High baseline, no upward drift yet.

2. **Datasets & software stay marginal**

   Data + software: 1,001 (2023) → 554 → 600 — barely 3–6% of output. FAIR data sharing lags open-access papers.

3. **Volume is flat, mix is the story**

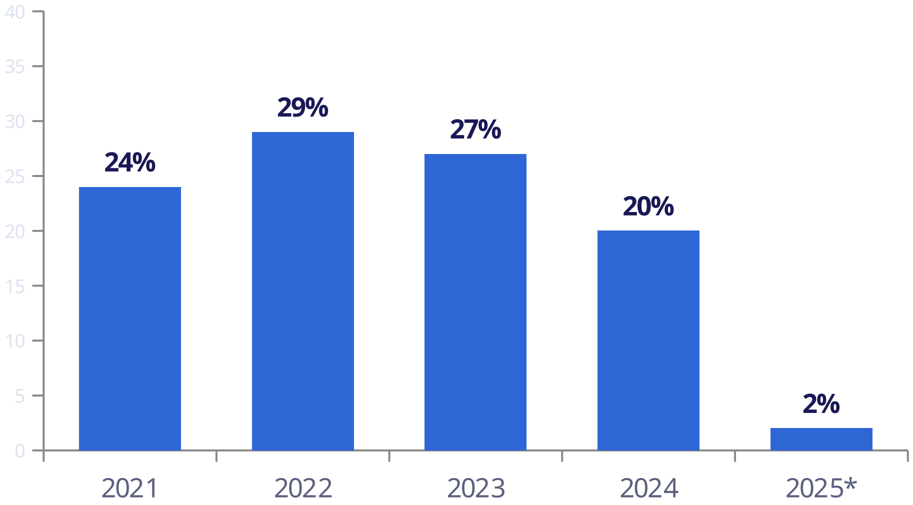
   Total output steady at ~16–17K/yr, so the signal for policy is composition — OA and data practice — not growth.

Same three queries, any institution: *an open, reproducible alternative to a proprietary OS dashboard.*

MONITORING · EXAMPLE 3

STEM × SSH integration via FoS co-occurrence

sample SSH-classified products per year, test each record's Fields-of-Science subjects for a co-occurring STEM code



Share of top-cited SSH work carrying a STEM FoS code

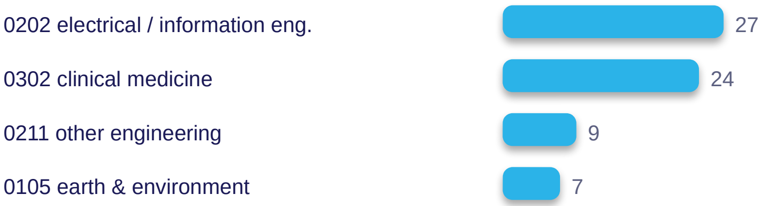
What Claude concludes

1 **Roughly 1 in 4 stays cross-cutting**

In high-impact SSH literature, STEM co-occurrence holds at ~20–29% across 2021–24 — stable, not rising.

2 **Top STEM partners of SSH**

Information engineering and clinical medicine lead — the AI-in-society and health-policy interfaces.



***2025 partial** *(classification lag). Sampled estimate over top-cited strata — directional, not a census; the query is fully reproducible.*

OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

TAKEAWAYS

A graph you can talk to

1. **Rich, typed model**
   Six entity types and 7B citation relationships turn scattered metadata into a navigable network.

2. **Open by design**
   Monthly releases, open APIs, Zenodo dumps — verifiable numbers, no vendor lock-in.

3. **Discovery in one hop**
   Influence-ranked search and citation traversal surface landmark and cross-domain work fast.

4. **Monitoring on tap**
   Topic trends, institutional open science, cross-disciplinary integration — repeatable and schedulable.

OpenAIRE
OPEN SCIENCE POLICY
TRAINING PROGRAMME

THANK YOU!

RESOURCES & CONTACTS

OpenAIRE Graph

graph.openaire.eu

OpenAIRE MONITOR

monitor.openaire.eu